

# Amarsh Pedapati

Data Scientist | AI Engineer | LLMs & Computer Vision

[in amارش](#) | [M amارش.pedapati](#) | [amارش](#) | [amarsh.pedapati@gmail.com](mailto:amarsh.pedapati@gmail.com) | [+91-9959822444](tel:+919959822444)

## EDUCATION

**Indian Institute of Technology, Hyderabad**

*Bachelor of Technology, Artificial Intelligence (GPA - 8.34/10)*

**Jul 2019 – May 2023**

*Hyderabad, India*

## WORK EXPERIENCE

**Data Scientist | Blend360, Hyderabad, India**

**Oct 2023 – Present**

**For Walmart**

- Led end-to-end customer segmentation and uplift modeling for a large-scale holiday sales campaign using **Random Forest** treatment and control models, achieving **45%**
- Built a high-performance Spark datamart (**7x** faster) and engineered ETL/preprocessing pipelines with SQL optimization and Python, achieving runtimes over **1000x** faster.
- Improved model lift by **100%**, applied **probability calibration**, and enabled statistically significant targeting of **13.4M** customers from a **130M** population through power analysis.

**For VISA**

- Developed an explainable AI web app in Streamlit enabling clients to explore model behavior via trendline discrepancy analysis, residual-based Shapley explainability, and forecast comparison.
- Engineered advanced LangChain reduction pipelines (multi-core custom implementation) to summarize discrepancies with LLMs, generating both technical insights and executive-level summaries.
- Integrated an LLM-powered chatbot with Agents, combining RAG (ChromaDB) for discrepancy-focused Q&A and data exploration via PandasAI, improving model interpretability and client trust.

**Data Scientist Intern | AIBOD, Fukuoka, Japan**

**Jan 2022 – June 2022**

- Enhanced reliability of an AI-powered object detection system for unmanned cash registers by resolving out-of-distribution (OOD) misclassification issues, which were a major blocker for retail automation deployment.
- Optimized classification thresholds through linear, polynomial, ensemble-based, and ML-driven calibration methods (logistic regression, gradient boosting), enhancing confidence scoring.
- Applied advanced data augmentation (synthetic generation, domain randomization, adversarial noise) to boost robustness across diverse real-world conditions.
- Integrated an OOD detection pipeline with uncertainty estimation and Mahalanobis distance, reducing false positives and achieving a **32%** drop in misclassification rate with an **18%** accuracy gain.

**Data Scientist Intern | ManaliSwing, Manali, India**

**Sept 2023 – Dec 2023**

- Designed and deployed a real-time Computer Vision-based safety verification system for adventure sports, detecting carabiner open/closed status to prevent safety failures.
- Built a diverse dataset across challenging environmental conditions, benchmarked state-of-the-art object detection models, and implemented YOLOv5 with live GoPro video streaming optimized for low-latency inference.
- Delivered high reliability in safety-critical detection (**95%** accuracy, **92%** precision, **93%** recall), and deployed the model in production at a commercial adventure facility in Manali, significantly strengthening operational safety and compliance standards.

## TECHNICAL SKILLS

**Programming & AI:** Python, C++, SQL | Machine Learning, Deep Learning | LLMs, Generative AI (RAG, Fine-tuning, Prompt Engineering, Agents) | Data Engineering

**Frameworks & Libraries:** PyTorch, Fastai, Hugging Face, LangChain, Streamlit

**Data Science & Visualization:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Plotly | Tableau, Power BI | Dashboard Design & Data Visualization Best Practices

**Cloud & MLOps:** AWS (SageMaker, Lambda), Azure ML, Databricks, MLflow, Docker, Kubernetes

**Databases & Storage:** Snowflake, PostgreSQL, MySQL | Vector DBs (FAISS, Pinecone, Weaviate)

**Certifications:** AWS Certified AI Practitioner, Oracle Generative AI Professional